

## ■ 시나리오 별 백업 복구(User Managed) - Archive log mode

### KEY WORDS

- <시나리오 9> 모든 데이터파일 손상되었을 경우 복구
- <시나리오 10> 모든 데이터파일 손상된 경우 새로운 위치로 복구
- <시나리오 11> undo data file 손상되었을 경우 복구
- <시나리오 12> 운영중에 undo data file 손상된 경우 복구
- <시나리오 13> 복구 작업시 필요한 archive log file 이 한쪽에 없을 경우

### <<시나리오 9>> 모든 데이터파일 손상되었을 경우 복구

```
SQL> !ls /home/oracle/backup/open_bkp/
```

```
bnr.sh control01.ctl control_20250509.ctl redo02.log something.lst system01.dbf undotbs01.dbf
bnr1.sh control02.ctl redo01.log redo03.log sysaux01.dbf temp01.dbf users01.dbf
```

```
SQL> select file#,name,status from v$datafile;
```

```

FILE#
-----
NAME
-----
STATUS
-----
          1
/u01/app/oracle/oradata/ORCL/system01.dbf
SYSTEM

          3
/u01/app/oracle/oradata/ORCL/sysaux01.dbf
ONLINE

FILE#
-----
NAME
-----
STATUS
-----

          4
/u01/app/oracle/oradata/ORCL/undotbs01.dbf
ONLINE
```

7

/u01/app/oracle/oradata/ORCL/users01.dbf

FILE#

-----

NAME

-----

STATUS

-----

ONLINE

SQL&gt; col name for a50

SQL&gt; /

FILE# NAME

STATUS

-----

|   |                                            |        |
|---|--------------------------------------------|--------|
| 1 | /u01/app/oracle/oradata/ORCL/system01.dbf  | SYSTEM |
| 3 | /u01/app/oracle/oradata/ORCL/sysaux01.dbf  | ONLINE |
| 4 | /u01/app/oracle/oradata/ORCL/undotbs01.dbf | ONLINE |
| 7 | /u01/app/oracle/oradata/ORCL/users01.dbf   | ONLINE |

SQL&gt; alter database begin backup;

Database altered.

```
SQL> !cp -v /u01/app/oracle/oradata/ORCL/*.* /home/oracle/backup/open_bkp/
'/u01/app/oracle/oradata/ORCL/bnr.sh' -> '/home/oracle/backup/open_bkp/bnr.sh'
'/u01/app/oracle/oradata/ORCL/bnr1.sh' -> '/home/oracle/backup/open_bkp/bnr1.sh'
'/u01/app/oracle/oradata/ORCL/control01.ctl' -> '/home/oracle/backup/open_bkp/control01.ctl'
'/u01/app/oracle/oradata/ORCL/control02.ctl' -> '/home/oracle/backup/open_bkp/control02.ctl'
'/u01/app/oracle/oradata/ORCL/redo01.log' -> '/home/oracle/backup/open_bkp/redo01.log'
'/u01/app/oracle/oradata/ORCL/redo02.log' -> '/home/oracle/backup/open_bkp/redo02.log'
'/u01/app/oracle/oradata/ORCL/redo03.log' -> '/home/oracle/backup/open_bkp/redo03.log'
'/u01/app/oracle/oradata/ORCL/something.lst' -> '/home/oracle/backup/open_bkp/something.lst'
'/u01/app/oracle/oradata/ORCL/sysaux01.dbf' -> '/home/oracle/backup/open_bkp/sysaux01.dbf'
'/u01/app/oracle/oradata/ORCL/system01.dbf' -> '/home/oracle/backup/open_bkp/system01.dbf'
'/u01/app/oracle/oradata/ORCL/temp01.dbf' -> '/home/oracle/backup/open_bkp/temp01.dbf'
'/u01/app/oracle/oradata/ORCL/undotbs01.dbf' -> '/home/oracle/backup/open_bkp/undotbs01.dbf'
'/u01/app/oracle/oradata/ORCL/users01.dbf' -> '/home/oracle/backup/open_bkp/users01.dbf'
```

```
SQL> select a.file#, a.name, a.checkpoint_change#, b.status, b.change#, to_char(b.time, 'yyyy-mm-dd
hh24:mi:ss.sssss') time from v$datafile a, v$backup b where a.file# = b.file#;
```

| FILE#  | NAME                                       | CHECKPOINT_CHANGE# |
|--------|--------------------------------------------|--------------------|
| 1      | /u01/app/oracle/oradata/ORCL/system01.dbf  | 2438528            |
| ACTIVE | 2438528 2025-05-12 11:06:14.39974          |                    |
| 3      | /u01/app/oracle/oradata/ORCL/sysaux01.dbf  | 2438528            |
| ACTIVE | 2438528 2025-05-12 11:06:14.39974          |                    |
| 4      | /u01/app/oracle/oradata/ORCL/undotbs01.dbf | 2438528            |
| ACTIVE | 2438528 2025-05-12 11:06:14.39974          |                    |

| FILE#  | NAME                                     | CHECKPOINT_CHANGE# |
|--------|------------------------------------------|--------------------|
| 7      | /u01/app/oracle/oradata/ORCL/users01.dbf | 2438528            |
| ACTIVE | 2438528 2025-05-12 11:06:14.39974        |                    |

```
SQL> !ls -al /home/oracle/bakcup/open_bkp/
ls: cannot access /home/oracle/bakcup/open_bkp/: No such file or directory
```

```
SQL> !ls -al /home/oracle/backup/open_bkp
total 2089104
drwxr-xr-x. 2 oracle dba      4096 May  9 15:37 .
drwxr-xr-x. 6 oracle dba       68 May  9 15:15 ..
-rw-r--r--. 1 oracle dba     342 May 12 11:06 bnr.sh
-rw-r--r--. 1 oracle dba     258 May 12 11:06 bnr1.sh
-rw-r-----. 1 oracle dba 10928128 May 12 11:06 control01.ctl
-rw-r-----. 1 oracle dba 10928128 May 12 11:06 control02.ctl
-rw-r-----. 1 oracle dba 10928128 May  9 15:37 control_20250509.ctl
-rw-r-----. 1 oracle dba 10486272 May 12 11:06 redo01.log
-rw-r-----. 1 oracle dba 10486272 May 12 11:06 redo02.log
-rw-r-----. 1 oracle dba 10486272 May 12 11:06 redo03.log
```

```
-rw-r--r--. 1 oracle dba      153 May 12 11:06 something.lst
-rw-r-----. 1 oracle dba 650125312 May 12 11:06 sysaux01.dbf
-rw-r-----. 1 oracle dba 943726592 May 12 11:06 system01.dbf
-rw-r-----. 1 oracle dba 125837312 May 12 11:06 temp01.dbf
-rw-r-----. 1 oracle dba 361766912 May 12 11:07 undotbs01.dbf
-rw-r-----. 1 oracle dba   5251072 May 12 11:07 users01.dbf
```

```
SQL> alter datatabase end backup;
alter datatabase end backup
```

\*

ERROR at line 1:

ORA-00940: invalid ALTER command

```
SQL> alter database end backup;
```

Database altered.

```
SQL> select a.file#, a.name, a.checkpoint_change#, b.status, b.change#, to_char(b.time, 'yyyy-mm-dd
hh24:mi:ss.sssss') time from v$datafile a, v$backup b where a.file# = b.file#;
```

| FILE#      | NAME                                       | CHECKPOINT_CHANGE# |
|------------|--------------------------------------------|--------------------|
| 1          | /u01/app/oracle/oradata/ORCL/system01.dbf  | 2438528            |
| NOT ACTIVE | 2438528 2025-05-12 11:06:14.39974          |                    |
| 3          | /u01/app/oracle/oradata/ORCL/sysaux01.dbf  | 2438528            |
| NOT ACTIVE | 2438528 2025-05-12 11:06:14.39974          |                    |
| 4          | /u01/app/oracle/oradata/ORCL/undotbs01.dbf | 2438528            |
| NOT ACTIVE | 2438528 2025-05-12 11:06:14.39974          |                    |

| FILE#      | NAME                                     | CHECKPOINT_CHANGE# |
|------------|------------------------------------------|--------------------|
| 7          | /u01/app/oracle/oradata/ORCL/users01.dbf | 2438528            |
| NOT ACTIVE | 2438528 2025-05-12 11:06:14.39974        |                    |

```
SQL> alter database backup controlfile to '/home/oracle/backup/open_bkp/control_20250512.ctl' ;
```

Database altered.

```
SQL> !ls /home/oracle/arch*
```

/home/oracle/arch1:

```
0001_0000000010_1200585133.arc      0001_0000000014_1200585133.arc      0001_0000000018_1200585133.arc
0001_0000000022_1200585133.arc
0001_0000000011_1200585133.arc      0001_0000000015_1200585133.arc      0001_0000000019_1200585133.arc
0001_0000000023_1200585133.arc
0001_0000000012_1200585133.arc      0001_0000000016_1200585133.arc      0001_0000000020_1200585133.arc
0001_0000000013_1200585133.arc      0001_0000000017_1200585133.arc      0001_0000000021_1200585133.arc
```

/home/oracle/arch2:

```
0001_0000000010_1200585133.arc      0001_0000000014_1200585133.arc      0001_0000000018_1200585133.arc
0001_0000000022_1200585133.arc
0001_0000000011_1200585133.arc      0001_0000000015_1200585133.arc      0001_0000000019_1200585133.arc
0001_0000000023_1200585133.arc
0001_0000000012_1200585133.arc      0001_0000000016_1200585133.arc      0001_0000000020_1200585133.arc
0001_0000000013_1200585133.arc      0001_0000000017_1200585133.arc      0001_0000000021_1200585133.arc
```

```
SQL> !rm -rf /home/oracle/arch{1,2}/*
```

```
SQL> !ls /home/oracle/arch*
```

/home/oracle/arch1:

/home/oracle/arch2:

```
SQL> alter system archive log current;
```

System altered.

```
SQL> !ls /home/oracle/arch*
```

/home/oracle/arch1:

```
0001_0000000024_1200585133.arc
```

/home/oracle/arch2:

```
0001_0000000024_1200585133.arc
```

```
SQL> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
```

b.next\_change# from v\$logfile a, v\$log b where a.group# = b.group# order by 1,2;

```
GROUP# SEQUENCE#
-----
MEMBER
-----
MB STATUS          ARC FIRST_CHANGE# NEXT_CHANGE#
-----
1          25
/u01/app/oracle/oradata/ORCL/redo01.log
10 CURRENT          NO          2439424    1.8447E+19

2          23
/u01/app/oracle/oradata/ORCL/redo02.log
10 INACTIVE          YES          2433771    2438204
```

```
GROUP# SEQUENCE#
-----
MEMBER
-----
MB STATUS          ARC FIRST_CHANGE# NEXT_CHANGE#
-----
3          24
/u01/app/oracle/oradata/ORCL/redo03.log
10 ACTIVE           YES          2438204    2439424
```

## &lt;&lt;시나리오 10&gt;&gt; 모든 데이터파일 손상되었을 경우 복구(새로운 위치)

```
SQL> select file#, name, checkpoint_change#, status from v$datafile;
```

| FILE# | NAME                                       | CHECKPOINT_CHANGE# | STATUS |
|-------|--------------------------------------------|--------------------|--------|
| 1     | /u01/app/oracle/oradata/ORCL/system01.dbf  | 2439424            | SYSTEM |
| 3     | /u01/app/oracle/oradata/ORCL/sysaux01.dbf  | 2439424            | ONLINE |
| 4     | /u01/app/oracle/oradata/ORCL/undotbs01.dbf | 2439424            | ONLINE |
| 7     | /u01/app/oracle/oradata/ORCL/users01.dbf   | 2439424            | ONLINE |

# 장애 발생

```
SQL> ! ls /u01/app/oracle/oradata/ORCL/*.dbf
```

```
/u01/app/oracle/oradata/ORCL/sysaux01.dbf
```

```
/u01/app/oracle/oradata/ORCL/temp01.dbf
```

```
/u01/app/oracle/oradata/ORCL/users01.dbf
```

```
/u01/app/oracle/oradata/ORCL/system01.dbf /u01/app/oracle/oradata/ORCL/undotbs01.dbf
```

```
SQL> ! rm /u01/app/oracle/oradata/ORCL/*.dbf
```

```
SQL> select name dbname, open_mode, log_mode, checkpoint_change# from v$database;
```

| DBNAME | OPEN_MODE  | LOG_MODE   | CHECKPOINT_CHANGE# |
|--------|------------|------------|--------------------|
| ORCL   | READ WRITE | ARCHIVELOG | 2443327            |

```
SQL> alter system checkpoint;
```

System altered.

```
SQL> conn / as sysdba
```

Connected.

```
SQL> ! ls /u01/app/oracle/oradata/ORCL/*.dbf
```

```
ls: cannot access /u01/app/oracle/oradata/ORCL/*.dbf: No such file or directory
```

-- Alert log file 확인하기

- 오류나지 않을 수 있으나 다른 세션으로 hr계정 접속시 오류발생

```
SQL> conn hr/hr
```

ERROR:

ORA-00604: error occurred at recursive SQL level 1

ORA-01116: error in opening database file 1  
 ORA-01110: data file 1: '/u01/app/oracle/oradata/ORCL/system01.dbf'  
 ORA-27041: unable to open file  
 Linux-x86\_64 Error: 2: No such file or directory  
 Additional information: 3  
 Warning: You are no longer connected to ORACLE.

- 접속 가능할 경우 shutdown abort

SQL> shut abort  
 ORACLE instance shut down.

SQL> startup  
 ORACLE instance started.  
 Total System Global Area 1811936368 bytes  
 Fixed Size 8897648 bytes  
 Variable Size 436207616 bytes  
 Database Buffers 1358954496 bytes  
 Redo Buffers 7876608 bytes  
 Database mounted.  
 ORA-01157: cannot identify/lock data file 1 - see DBWR trace file  
 ORA-01110: data file 1: '/u01/app/oracle/oradata/ORCL/system01.dbf'

SQL> select \* from v\$recover\_file;

| FILE# | ONLINE | ONLINE_ ERROR | CHANGE#        | TIME | CON_ID |
|-------|--------|---------------|----------------|------|--------|
| 1     | ONLINE | ONLINE        | FILE NOT FOUND | 0    | 0      |
| 3     | ONLINE | ONLINE        | FILE NOT FOUND | 0    | 0      |
| 4     | ONLINE | ONLINE        | FILE NOT FOUND | 0    | 0      |
| 7     | ONLINE | ONLINE        | FILE NOT FOUND | 0    | 0      |

SQL> !ls /u01/app/oracle/oradata/ORCL  
 bnr.sh bnr1.sh control01.ctl control02.ctl redo01.log redo02.log redo03.log something.lst

SQL> select status from v\$instance;

STATUS  
 -----  
 MOUNTED

#Recovery

### 1. 최근 백업본 찾아 restore

```
SQL> !ls /home/oracle/backup/open_bkp/*.dbf
/home/oracle/backup/open_bkp/sysaux01.dbf          /home/oracle/backup/open_bkp/temp01.dbf
/home/oracle/backup/open_bkp/users01.dbf
/home/oracle/backup/open_bkp/system01.dbf  /home/oracle/backup/open_bkp/undotbs01.dbf
```

```
SQL> !cp -v /home/oracle/backup/open_bkp/*.dbf /u01/app/oracle/oradata/ORCL/
'/home/oracle/backup/open_bkp/sysaux01.dbf' -> '/u01/app/oracle/oradata/ORCL/sysaux01.dbf'
'/home/oracle/backup/open_bkp/system01.dbf' -> '/u01/app/oracle/oradata/ORCL/system01.dbf'
'/home/oracle/backup/open_bkp/temp01.dbf' -> '/u01/app/oracle/oradata/ORCL/temp01.dbf'
'/home/oracle/backup/open_bkp/undotbs01.dbf' -> '/u01/app/oracle/oradata/ORCL/undotbs01.dbf'
'/home/oracle/backup/open_bkp/users01.dbf' -> '/u01/app/oracle/oradata/ORCL/users01.dbf'
```

### 2. 백업 이후 변경정보(리두) 적용

```
SQL> recover database;
Media recovery complete.
```

- 리두만 적용하면 끝이라면 화면에 표기되지 않는다.

- Alert log file

2025-05-09T16:47:15.197369+09:00

Recovery of Online Redo Log: Thread 1 Group 1 Seq 19 Reading mem 0

Mem# 0: /u01/app/oracle/oradata/ORCL/redo01.log

2025-05-09T16:47:15.202756+09:00

Recovery of Online Redo Log: Thread 1 Group 2 Seq 20 Reading mem 0

Mem# 0: /u01/app/oracle/oradata/ORCL/redo02.log

2025-05-09T16:47:15.205594+09:00

Recovery of Online Redo Log: Thread 1 Group 3 Seq 21 Reading mem 0

Mem# 0: /u01/app/oracle/oradata/ORCL/redo03.log

2025-05-09T16:47:15.376615+09:00

Media Recovery Complete (orcl)

Completed: ALTER DATABASE RECOVER CONTINUE DEFAULT

### 3. 데이터베이스 open

```
SQL> alter database open;
```

Database altered.

```
SQL> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a,
v$tablespace b where a.ts# = b.ts#;
```

| FILE#              | TBS_NAME | FILE_NAME                                  |
|--------------------|----------|--------------------------------------------|
| CHECKPOINT_CHANGE# | STATUS   |                                            |
| -----              |          |                                            |
| 1                  | SYSTEM   | /u01/app/oracle/oradata/ORCL/system01.dbf  |
| 2646238            | SYSTEM   |                                            |
| 3                  | SYSAUX   | /u01/app/oracle/oradata/ORCL/sysaux01.dbf  |
| 2646238            | ONLINE   |                                            |
| 4                  | UNDOTBS1 | /u01/app/oracle/oradata/ORCL/undotbs01.dbf |
| 2646238            | ONLINE   |                                            |
| 7                  | USERS    | /u01/app/oracle/oradata/ORCL/users01.dbf   |
| 2646238            | ONLINE   |                                            |

```
SQL> select count(*) from hr.employees;
```

| COUNT(*) |
|----------|
| -----    |
| 107      |

## &lt;&lt;시나리오 11&gt;&gt; undo data file 손상되었을 경우 복구

```
SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a,
v$tablespace b where a.ts# = b.ts#;
```

| FILE# | TBS_NAME | FILE_NAME                                  | CHECKPOINT_CHANGE# | STATUS |
|-------|----------|--------------------------------------------|--------------------|--------|
| 1     | SYSTEM   | /u01/app/oracle/oradata/ORCL/system01.dbf  | 2192042            | SYSTEM |
| 3     | SYSAUX   | /u01/app/oracle/oradata/ORCL/sysaux01.dbf  | 2192042            | ONLINE |
| 4     | UNDOTBS1 | /u01/app/oracle/oradata/ORCL/undotbs01.dbf | 2192042            | ONLINE |
| 7     | USERS    | /u01/app/oracle/oradata/ORCL/users01.dbf   | 2192042            | ONLINE |

# 장애 발생

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/*.dbf
/u01/app/oracle/oradata/ORCL/sysaux01.dbf /u01/app/oracle/oradata/ORCL/undotbs01.dbf
/u01/app/oracle/oradata/ORCL/system01.dbf /u01/app/oracle/oradata/ORCL/users01.dbf
/u01/app/oracle/oradata/ORCL/temp01.dbf
```

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/*.dbf
```

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/*.dbf
ls: cannot access /u01/app/oracle/oradata/ORCL/*.dbf: No such file or directory
```

```
SYS@orcl> alter system checkpoint;
```

```
alter system checkpoint
```

```
*
```

```
ERROR at line 1:
```

```
ORA-03113: end-of-file on communication channel
```

```
Process ID: 18079
```

```
Session ID: 497 Serial number: 28716
```

```
SYS@orcl> conn / as sysdba
```

```
Connected to an idle instance.
```

```
SYS@orcl> startup
```

```
ORACLE instance started.
```

Total System Global Area 1811936368 bytes  
Fixed Size 8897648 bytes  
Variable Size 436207616 bytes  
Database Buffers 1358954496 bytes  
Redo Buffers 7876608 bytes

Database mounted.

ORA-01157: cannot identify/lock data file 1 - see DBWR trace file

ORA-01110: data file 1: '/u01/app/oracle/oradata/ORCL/system01.dbf'

-Alert log file

2025-05-12T18:58:31.070781+09:00

Errors in file /u01/app/oracle/diag/rdbms/orcl/orcl/trace/orcl\_ora\_19616.trc:

ORA-01157: cannot identify/lock data file 1 - see DBWR trace file

ORA-01110: data file 1: '/u01/app/oracle/oradata/ORCL/system01.dbf'

2025-05-12T18:58:31.070963+09:00

Errors in file /u01/app/oracle/diag/rdbms/orcl/orcl/trace/orcl\_ora\_19616.trc:

ORA-01157: cannot identify/lock data file 1 - see DBWR trace file

ORA-01110: data file 1: '/u01/app/oracle/oradata/ORCL/system01.dbf'

ORA-1157 signalled during: ALTER DATABASE OPEN...

2025-05-12T18:58:31.094759+09:00

Errors in file /u01/app/oracle/diag/rdbms/orcl/orcl/trace/orcl\_mz00\_19618.trc:

ORA-01110: data file 3: '/u01/app/oracle/oradata/ORCL/sysaux01.dbf'

ORA-01565: error in identifying file '/u01/app/oracle/oradata/ORCL/sysaux01.dbf'

ORA-27037: unable to obtain file status

Linux-x86\_64 Error: 2: No such file or directory

Additional information: 7

2025-05-12T18:58:31.149857+09:00

Errors in file /u01/app/oracle/diag/rdbms/orcl/orcl/trace/orcl\_mz00\_19618.trc:

ORA-01110: data file 4: '/u01/app/oracle/oradata/ORCL/undotbs01.dbf'

ORA-01565: error in identifying file '/u01/app/oracle/oradata/ORCL/undotbs01.dbf'

ORA-27037: unable to obtain file status

Linux-x86\_64 Error: 2: No such file or directory

Additional information: 7

2025-05-12T18:58:31.254561+09:00

Errors in file /u01/app/oracle/diag/rdbms/orcl/orcl/trace/orcl\_mz00\_19618.trc:

ORA-01110: data file 7: '/u01/app/oracle/oradata/ORCL/users01.dbf'

ORA-01565: error in identifying file '/u01/app/oracle/oradata/ORCL/users01.dbf'

ORA-27037: unable to obtain file status

Linux-x86\_64 Error: 2: No such file or directory

Additional information: 7

SYS@orcl> select \* from v\$recover\_file;

| FILE# | ONLINE | ONLINE_ ERROR | CHANGE#        | TIME | CON_ID |
|-------|--------|---------------|----------------|------|--------|
| 1     | ONLINE | ONLINE        | FILE NOT FOUND | 0    | 0      |
| 3     | ONLINE | ONLINE        | FILE NOT FOUND | 0    | 0      |
| 4     | ONLINE | ONLINE        | FILE NOT FOUND | 0    | 0      |
| 7     | ONLINE | ONLINE        | FILE NOT FOUND | 0    | 0      |

SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/  
control01.ctl control02.ctl initorcl.ora redo01.log redo02.log redo03.log

SYS@orcl> select status from v\$instance;

STATUS  
-----  
MOUNTED

SYS@orcl> select a.file#, b.name tbs\_name, a.name file\_name, a.checkpoint\_change#, a.status from v\$datafile a, v\$tablespace b where a.ts# = b.ts#;

| FILE# | TBS_NAME | FILE_NAME                                  | CHECKPOINT_CHANGE# |
|-------|----------|--------------------------------------------|--------------------|
| 1     | SYSTEM   | /u01/app/oracle/oradata/ORCL/system01.dbf  | 2192042            |
| 3     | SYSAUX   | /u01/app/oracle/oradata/ORCL/sysaux01.dbf  | 2192042            |
| 4     | UNDOTBS1 | /u01/app/oracle/oradata/ORCL/undotbs01.dbf | 2192042            |
| 7     | USERS    | /u01/app/oracle/oradata/ORCL/users01.dbf   | 2192042            |

# Recovery

## 1. 최근 백업본 찾아서 새로운 위치로 restore

```
SYS@orcl> ! ls /home/oracle/backup/open_bkp/*.dbf
```

```
/home/oracle/backup/open_bkp/sysaux01.dbf /home/oracle/backup/open_bkp/undotbs01.dbf
```

```
/home/oracle/backup/open_bkp/system01.dbf /home/oracle/backup/open_bkp/users01.dbf
```

```
/home/oracle/backup/open_bkp/temp01.dbf
```

```
SYS@orcl> ! cp -v /home/oracle/backup/open_bkp/*.dbf /home/oracle/oradata/
```

```
'/home/oracle/backup/open_bkp/sysaux01.dbf' -> '/home/oracle/oradata/sysaux01.dbf'
```

```
'/home/oracle/backup/open_bkp/system01.dbf' -> '/home/oracle/oradata/system01.dbf'
```

```
'/home/oracle/backup/open_bkp/temp01.dbf' -> '/home/oracle/oradata/temp01.dbf'
```

```
'/home/oracle/backup/open_bkp/undotbs01.dbf' -> '/home/oracle/oradata/undotbs01.dbf'
```

```
'/home/oracle/backup/open_bkp/users01.dbf' -> '/home/oracle/oradata/users01.dbf'
```

```
SYS@orcl> alter database rename file '/u01/app/oracle/oradata/ORCL/system01.dbf' to
'/home/oracle/oradata/system01.dbf';
```

Database altered.

```
SYS@orcl> alter database rename file '/u01/app/oracle/oradata/ORCL/sysaux01.dbf' to
'/home/oracle/oradata/sysaux01.dbf';
```

Database altered.

```
SYS@orcl> alter database rename file '/u01/app/oracle/oradata/ORCL/undotbs01.dbf' to
'/home/oracle/oradata/undotbs01.dbf';
```

Database altered.

```
SYS@orcl> alter database rename file '/u01/app/oracle/oradata/ORCL/users01.dbf' to
'/home/oracle/oradata/users01.dbf';
```

Database altered.

```
SYS@orcl> alter database rename file '/u01/app/oracle/oradata/ORCL/temp01.dbf' to
'/home/oracle/oradata/temp01.dbf';
```

Database altered.

```
SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a,
v$tablespace b where a.ts# = b.ts#;
```

| FILE# | TBS_NAME | FILE_NAME                         | CHECKPOINT_CHANGE# |
|-------|----------|-----------------------------------|--------------------|
| 1     | SYSTEM   | /home/oracle/oradata/system01.dbf | 2192042            |

SYSTEM

|   |         |                                   |         |
|---|---------|-----------------------------------|---------|
| 3 | SYS_AUX | /home/oracle/oradata/sysaux01.dbf | 2192042 |
|---|---------|-----------------------------------|---------|

ONLINE

|   |          |                                    |         |        |
|---|----------|------------------------------------|---------|--------|
| 4 | UNDOTBS1 | /home/oracle/oradata/undotbs01.dbf | 2192042 | ONLINE |
| 7 | USERS    | /home/oracle/oradata/users01.dbf   | 2192042 |        |

ONLINE

SYS@orcl> select name from v\$tempfile;

NAME

-----  
-----

/home/oracle/oradata/temp01.dbf

### 3. 백업 이후 변경정보(리두) 적용

SYS@orcl> recover database

Media recovery complete.

-- Alert log file

2025-05-12T19:13:15.904323+09:00

ALTER DATABASE RECOVER database

2025-05-12T19:13:15.904517+09:00

Media Recovery Start

Started logmerger process

2025-05-12T19:13:16.160736+09:00

Parallel Media Recovery started with 8 slaves

2025-05-12T19:13:16.215349+09:00

Recovery of Online Redo Log: Thread 1 Group 2 Seq 62 Reading mem 0

Mem# 0: /u01/app/oracle/oradata/ORCL/redo02.log

2025-05-12T19:13:16.603468+09:00

Media Recovery Complete (orcl)

Completed: ALTER DATABASE RECOVER database

### 4. 데이터베이스 open

SYS@orcl> alter database open;

Database altered.

SYS@orcl> select a.file#, b.name tbs\_name, a.name file\_name, a.checkpoint\_change#, a.status from v\$datafile a,  
v\$tablespace b where a.ts# = b.ts#;

| FILE#  | TBS_NAME | FILE_NAME                          | CHECKPOINT_CHANGE# |
|--------|----------|------------------------------------|--------------------|
| STATUS |          |                                    |                    |
| -----  |          |                                    |                    |
| 1      | SYSTEM   | /home/oracle/oradata/system01.dbf  | 2292380            |
| SYSTEM |          |                                    |                    |
| 3      | SYSAUX   | /home/oracle/oradata/sysaux01.dbf  | 2292380            |
| ONLINE |          |                                    |                    |
| 4      | UNDOTBS1 | /home/oracle/oradata/undotbs01.dbf | 2292380            |
| ONLINE |          |                                    |                    |
| 7      | USERS    | /home/oracle/oradata/users01.dbf   | 2292380            |
| ONLINE |          |                                    |                    |

# 데이터파일 이관작업(시나리오10 원래대로 돌아가기)

/home/oracle/oradata/\*.dbf -> /u01/app/oracle/oradata/ORCL/

### 1. 데이터베이스 정상적인 종료

```
SYS@orcl> shutdown immediate
Database closed.
Database dismounted.
ORACLE instance shut down.
```

### 2. 기존 파일을 새로운 위치로 move

```
SYS@orcl> ! mv /home/oracle/oradata/*.dbf /u01/app/oracle/oradata/ORCL/
```

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/
control01.ctl  initorcl.ora  redo02.log    sysaux01.dbf  temp01.dbf    users01.dbf
control02.ctl  redo01.log    redo03.log    system01.dbf  undotbs01.dbf
```

### 3. 데이터베이스 mount

```
SYS@orcl> startup mount
ORACLE instance started.
```

```
Total System Global Area 1543500144 bytes
Fixed Size                  8896880 bytes
```

Variable Size                    905969664 bytes  
 Database Buffers                620756992 bytes  
 Redo Buffers                    7876608 bytes  
 Database mounted.

#### 4. 컨트롤파일에 데이터파일 정보 변경

# alter database rename file '기존위치' to '새로운위치'

```
SYS@orcl> alter database rename file '/home/oracle/oradata/system01.dbf' to
'/u01/app/oracle/oradata/ORCL/system01.dbf';
Database altered.
```

```
SYS@orcl> alter database rename file '/home/oracle/oradata/sysaux01.dbf' to
'/u01/app/oracle/oradata/ORCL/sysaux01.dbf';
Database altered.
```

```
SYS@orcl> alter database rename file '/home/oracle/oradata/undotbs01.dbf' to
'/u01/app/oracle/oradata/ORCL/undotbs01.dbf';
Database altered.
```

```
SYS@orcl> alter database rename file '/home/oracle/oradata/users01.dbf' to
'/u01/app/oracle/oradata/ORCL/users01.dbf';
Database altered.
```

```
SYS@orcl> alter database rename file '/home/oracle/oradata/temp01.dbf' to
'/u01/app/oracle/oradata/ORCL/temp01.dbf';
Database altered.
```

```
SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a,
v$tablespace b where a.ts# = b.ts#;
```

| FILE# | TBS_NAME | FILE_NAME                                  | CHECKPOINT_CHANGE# |
|-------|----------|--------------------------------------------|--------------------|
| 1     | SYSTEM   | /u01/app/oracle/oradata/ORCL/system01.dbf  | 2393443            |
| 3     | SYS_AUX  | /u01/app/oracle/oradata/ORCL/sysaux01.dbf  | 2393443            |
| 4     | UNDOTBS1 | /u01/app/oracle/oradata/ORCL/undotbs01.dbf | 2393443            |

|         |                                          |         |
|---------|------------------------------------------|---------|
| 7 USERS | /u01/app/oracle/oradata/ORCL/users01.dbf | 2393443 |
|---------|------------------------------------------|---------|

ONLINE

SYS@orcl> select name from v\$tempfile;

NAME

-----  
-----

/u01/app/oracle/oradata/ORCL/temp01.dbf

SYS@orcl> SYS@orcl> alter database open;  
Database altered.

## &lt;&lt;시나리오 12&gt;&gt; undo data file 손상되었을 경우 복구

SYS@ora19c> select segment\_id, segment\_name, owner, tablespace\_name, status from dba\_rollback\_segs;

| SEGMENT_ID | SEGMENT_NAME           | OWNER  | TABLESPACE_NAME | STATUS |
|------------|------------------------|--------|-----------------|--------|
| 0          | SYSTEM                 | SYS    | SYSTEM          | ONLINE |
| 1          | _SYSSMU1_1230671320\$  | PUBLIC | UNDOTBS1        | ONLINE |
| 2          | _SYSSMU2_3151764486\$  | PUBLIC | UNDOTBS1        | ONLINE |
| 3          | _SYSSMU3_1295854473\$  | PUBLIC | UNDOTBS1        | ONLINE |
| 4          | _SYSSMU4_1947518621\$  | PUBLIC | UNDOTBS1        | ONLINE |
| 5          | _SYSSMU5_1526312829\$  | PUBLIC | UNDOTBS1        | ONLINE |
| 6          | _SYSSMU6_2680029761\$  | PUBLIC | UNDOTBS1        | ONLINE |
| 7          | _SYSSMU7_2189229235\$  | PUBLIC | UNDOTBS1        | ONLINE |
| 8          | _SYSSMU8_4078897598\$  | PUBLIC | UNDOTBS1        | ONLINE |
| 9          | _SYSSMU9_2100413762\$  | PUBLIC | UNDOTBS1        | ONLINE |
| 10         | _SYSSMU10_2605481461\$ | PUBLIC | UNDOTBS1        | ONLINE |

11 rows selected.

<<hr>>

HR@orcl> update employees

2 set salary=4000

3 where employee\_id=101;

1 row updated.

<<sys>>

SYS@orcl> select s.username, s.sid, s.serial#, t.xidusn undo\_seg\_num, r.name undo\_seg\_name, t.ubafil  
undo\_datafile\_num, t.ubablk undo\_block, t.used\_ublk from v\$session s, v\$transaction t, v\$rollname r where s.taddr  
= t.addr and t.xidusn = r.usn;

| USERNAME | SID | SERIAL# | UNDO_SEG_NUM | UNDO_SEG_NAME         |
|----------|-----|---------|--------------|-----------------------|
| HR       | 252 | 9598    | 7            | _SYSSMU7_2329891355\$ |

  

| UNDO_DATAFILE_NUM | UNDO_BLOCK | USED_UBLK |
|-------------------|------------|-----------|
| 4                 | 231        | 1         |

# session kill

```
SYS@orcl> alter system kill session '252,9598' immediate;  
System altered.
```

<<hr>>

```
HR@orcl> select count(*) from employees;  
select count(*) from employees  
          *
```

```
ERROR at line 1:  
ORA-03135: connection lost contact  
Process ID: 22612  
Session ID: 252 Serial number: 9598
```

<<sys>>

# 장애 발생

```
SYS@orcl> SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/undotbs01.dbf  
/u01/app/oracle/oradata/ORCL/undotbs01.dbf
```

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/undotbs01.dbf
```

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/undotbs01.dbf  
ls: cannot access /u01/app/oracle/oradata/ORCL/undotbs01.dbf: No such file or directory
```

```
SYS@orcl> shut abort  
ORACLE instance shut down.
```

```
SYS@orcl> startup  
ORACLE instance started.
```

```
Total System Global Area 1811936368 bytes  
Fixed Size                  8897648 bytes  
Variable Size              436207616 bytes  
Database Buffers          1358954496 bytes  
Redo Buffers               7876608 bytes  
Database mounted.  
ORA-01157: cannot identify/lock data file 4 - see DBWR trace file
```

ORA-01110: data file 4: '/u01/app/oracle/oradata/ORCL/undotbs01.dbf'

SYS@ora19c> select \* from v\$recover\_file;

| FILE#        | ONLINE | ONLINE_ | ERROR          |
|--------------|--------|---------|----------------|
| CHANGE# TIME | CON_ID |         |                |
| -----        | -----  | -----   | -----          |
| 2            | ONLINE | ONLINE  | FILE NOT FOUND |
| 0            | 0      |         |                |

# Recovery

1. 최근 백업본 찾아 restore

SYS@orcl> ! ls /home/oracle/backup/open\_bkp/undotbs01.dbf /home/oracle/backup/open\_bkp/undotbs01.dbf

SYS@ora19c> SYS@orcl> !cp -v /home/oracle/backup/open\_bkp/undotbs01.dbf /u01/app/oracle/oradata/ORCL/  
'/home/oracle/backup/open\_bkp/undotbs01.dbf' -> '/u01/app/oracle/oradata/ORCL/undotbs01.dbf'

2. 백업 이후 변경정보(리두) 적용

SYS@orcl> recover tablespace undotbs1

Media recovery complete.

-- Alert log file

2025-05-12T19:40:34.011814+09:00

ALTER DATABASE RECOVER tablespace undotbs1

2025-05-12T19:40:34.012168+09:00

Media Recovery Start

2025-05-12T19:40:34.012583+09:00

Serial Media Recovery started

2025-05-12T19:40:34.043135+09:00

Recovery of Online Redo Log: Thread 1 Group 2 Seq 62 Reading mem 0

Mem# 0: /u01/app/oracle/oradata/ORCL/redo02.log

2025-05-12T19:40:34.049184+09:00

Recovery of Online Redo Log: Thread 1 Group 3 Seq 63 Reading mem 0

Mem# 0: /u01/app/oracle/oradata/ORCL/redo03.log

2025-05-12T19:40:34.093038+09:00

Media Recovery Complete (orcl)

Completed: ALTER DATABASE RECOVER tablespace undotbs1

## 3. 데이터베이스 open

```
SYS@orcl> alter database open;
Database altered.
```

&lt;&lt;hr&gt;&gt;

```
HR@orcl> conn hr/hr
Connected.
```

```
HR@orcl> update hr.employees set salary = salary * 1.1 where employee_id = 100;
```

&lt;&lt;sys&gt;&gt;

```
SYS@orcl> select s.username, s.sid, s.serial#, t.xidusn undo_seg_num, r.name undo_seg_name, t.ubafil
undo_datafile_num, t.ubablk undo_block, t.used_ublk from v$session s, v$transaction t, v$rollname r where s.taddr
= t.addr and t.xidusn = r.usn;
```

| USERNAME          |            |           |   | SID | SERIAL# | UNDO_SEG_NUM | UNDO_SEG_NAME        |
|-------------------|------------|-----------|---|-----|---------|--------------|----------------------|
| -----             |            |           |   |     |         |              |                      |
| - -----           |            |           |   |     |         |              |                      |
| UNDO_DATAFILE_NUM | UNDO_BLOCK | USED_UBLK |   |     |         |              |                      |
| -----             |            |           |   |     |         |              |                      |
| HR                |            |           |   | 10  | 53917   | 6            | _SYSSMU6_813816332\$ |
|                   | 4          | 1428      | 1 |     |         |              |                      |

&lt;&lt;hr&gt;&gt;

```
HR@orcl> rollback;
Rollback complete.
```

&lt;&lt;sys&gt;&gt;

```
SYS@orcl> select s.username, s.sid, s.serial#, t.xidusn undo_seg_num, r.name undo_seg_name, t.ubafil
undo_datafile_num, t.ubablk undo_block, t.used_ublk from v$session s, v$transaction t, v$rollname r where s.taddr
= t.addr and t.xidusn = r.usn;
```

no rows selected

## &lt;&lt;시나리오12&gt;&gt; 운영중에 undo data file 손상된 경우 복구

```
SYS@orcl> alter system switch logfile;
```

```
System altered.
```

```
SYS@orcl> /=
```

```
System altered.
```

```
SYS@orcl> /
```

```
System altered.
```

```
SYS@orcl>
```

```
System altered.
```

```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

| GROUP#      | SEQUENCE#  | MEMBER                                  | MB           |
|-------------|------------|-----------------------------------------|--------------|
| STATUS      | ARC        | FIRST_CHANGE#                           | NEXT_CHANGE# |
| -----       | -----      | -----                                   | -----        |
| 1           | 67         | /u01/app/oracle/oradata/ORCL/redo01.log | 20 INACTIVE  |
| YES 2595387 | 2595390    |                                         |              |
| 2           | 68         | /u01/app/oracle/oradata/ORCL/redo02.log | 20 CURRENT   |
| NO 2595390  | 1.8447E+19 |                                         |              |
| 3           | 66         | /u01/app/oracle/oradata/ORCL/redo03.log | 20 INACTIVE  |
| YES 2595380 | 2595387    |                                         |              |

```
SYS@orcl> select segment_id, segment_name, owner, tablespace_name, status from dba_rollback_segs;
```

| SEGMENT_ID | SEGMENT_NAME          | OWNER  | TABLESPACE_NAME | STATUS |
|------------|-----------------------|--------|-----------------|--------|
| 0          | SYSTEM                | SYS    | SYSTEM          | ONLINE |
| 1          | _SYSSMU1_1261223759\$ | PUBLIC | UNDOTBS1        | ONLINE |
| 2          | _SYSSMU2_27624015\$   | PUBLIC | UNDOTBS1        | ONLINE |
| 3          | _SYSSMU3_2421748942\$ | PUBLIC | UNDOTBS1        | ONLINE |
| 4          | _SYSSMU4_625702278\$  |        | PUBLIC UNDOTBS1 | ONLINE |
| 5          | _SYSSMU5_2101348960\$ | PUBLIC | UNDOTBS1        | ONLINE |
| 6          | _SYSSMU6_813816332\$  |        | PUBLIC UNDOTBS1 | ONLINE |
| 7          | _SYSSMU7_2329891355\$ | PUBLIC | UNDOTBS1        | ONLINE |
| 8          | _SYSSMU8_399776867\$  |        | PUBLIC UNDOTBS1 | ONLINE |
| 9          | _SYSSMU9_1692468413\$ | PUBLIC | UNDOTBS1        | ONLINE |

10\_SYSSMU10\_930580995\$

PUBLIC UNDOTBS1

ONLINE

11 rows selected.

&lt;&lt;hr&gt;&gt;

HR@orcl&gt; update hr.employees set salary = salary \* 1.1 where employee\_id = 100;

1 row updated.

HR@orcl&gt; select salary from hr.employees where employee\_id = 100;

SALARY

-----

26400

&lt;&lt;sys&gt;&gt;

SYS@orcl> select s.username, s.sid, s.serial#, t.xidusn undo\_seg\_num, r.name undo\_seg\_name, t.ubafil  
undo\_datafile\_num, t.ubablk undo\_block, t.used\_ublk from v\$session s, v\$transaction t, v\$rollname r where s.taddr  
= t.addr and t.xidusn = r.usn;

| USERNAME | SID | SERIAL# | UNDO_SEG_NUM | UNDO_SEG_NAME |
|----------|-----|---------|--------------|---------------|
|----------|-----|---------|--------------|---------------|

-----

- - - - -

| UNDO_DATAFILE_NUM | UNDO_BLOCK | USED_UBLK |
|-------------------|------------|-----------|
|-------------------|------------|-----------|

-----

|                       |       |       |   |
|-----------------------|-------|-------|---|
| HR                    | 10    | 53917 | 5 |
| _SYSSMU5_2101348960\$ |       |       |   |
| 4                     | 24325 | 1     |   |

# 장애 발생

SYS@orcl> SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/undotbs01.dbf  
/u01/app/oracle/oradata/ORCL/undotbs01.dbf

SYS@orcl&gt; ! rm /u01/app/oracle/oradata/ORCL/undotbs01.dbf

SYS@orcl&gt; ! ls /u01/app/oracle/oradata/ORCL/undotbs01.dbf

ls: cannot access /u01/app/oracle/oradata/ORCL/undotbs01.dbf: No such file or directory

&lt;&lt; hr &gt;&gt;

HR@orcl&gt; select salary from hr.employees where employee\_id = 200;

SALARY

-----  
4400

<<새로운 hr>>

[oracle@oel7v9r2 oradata]\$ sqlplus hr/hr

ERROR:

ORA-01116: error in opening database file 4

ORA-01110: data file 4: '/u01/app/oracle/oradata/ORCL/undotbs01.dbf'

ORA-27041: unable to open file

Linux-x86\_64 Error: 2: No such file or directory

Additional information: 3

-Alert log file 확인(시간이 지난후)

025-05-12T19:56:47.847570+09:00

Errors in file /u01/app/oracle/diag/rdbms/orcl/orcl/trace/orcl\_mz00\_24517.trc:

ORA-01110: data file 4: '/u01/app/oracle/oradata/ORCL/undotbs01.dbf'

ORA-01565: error in identifying file '/u01/app/oracle/oradata/ORCL/undotbs01.dbf'

ORA-27037: unable to obtain file status

Linux-x86\_64 Error: 2: No such file or directory

Additional information: 7

<<sys>>

SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/undotbs01.dbf

ls: cannot access /u01/app/oracle/oradata/ORCL/undotbs01.dbf: No such file or directory

# 새로운 undo tablespace 생성

SYS@orcl> create undo tablespace undo datafile '/u01/app/oracle/oradata/ORCL/undo01.dbf' size 50m  
autoextend on;

Tablespace created.

SYS@orcl> select file#, name, checkpoint\_change#, status from v\$datafile;

| FILE# | NAME | CHECKPOINT_CHANGE# | STATUS |
|-------|------|--------------------|--------|
|-------|------|--------------------|--------|

|   |                                            |         |        |
|---|--------------------------------------------|---------|--------|
| 1 | /u01/app/oracle/oradata/ORCL/system01.dbf  | 2595390 | SYSTEM |
| 3 | /u01/app/oracle/oradata/ORCL/sysaux01.dbf  | 2595390 | ONLINE |
| 4 | /u01/app/oracle/oradata/ORCL/undotbs01.dbf | 2595390 | ONLINE |
| 5 | /u01/app/oracle/oradata/ORCL/undo01.dbf    | 2596158 | ONLINE |
| 7 | /u01/app/oracle/oradata/ORCL/users01.dbf   | 2595390 | ONLINE |

SYS@orcl> select segment\_id, segment\_name, owner, tablespace\_name, status from dba\_rollback\_segs;

| SEGMENT_ID | SEGMENT_NAME          | OWNER  | TABLESPACE_NAME | STATUS |
|------------|-----------------------|--------|-----------------|--------|
| 0          | SYSTEM                | SYS    | SYSTEM          | ONLINE |
| 1          | _SYSSMU1_1261223759\$ | PUBLIC | UNDOTBS1        | ONLINE |
| 2          | _SYSSMU2_27624015\$   | PUBLIC | UNDOTBS1        | ONLINE |
| 3          | _SYSSMU3_2421748942\$ | PUBLIC | UNDOTBS1        | ONLINE |
| 4          | _SYSSMU4_625702278\$  |        | PUBLIC UNDOTBS1 | ONLINE |
| 5          | _SYSSMU5_2101348960\$ | PUBLIC | UNDOTBS1        | ONLINE |
| 6          | _SYSSMU6_813816332\$  |        | PUBLIC UNDOTBS1 | ONLINE |
| 7          | _SYSSMU7_2329891355\$ | PUBLIC | UNDOTBS1        | ONLINE |
| 8          | _SYSSMU8_399776867\$  |        | PUBLIC UNDOTBS1 | ONLINE |
| 9          | _SYSSMU9_1692468413\$ | PUBLIC | UNDOTBS1        | ONLINE |
| 10         | _SYSSMU10_930580995\$ | PUBLIC | UNDOTBS1        | ONLINE |

| SEGMENT_ID | SEGMENT_NAME           | OWNER  | TABLESPACE_NAME | STATUS  |
|------------|------------------------|--------|-----------------|---------|
| 11         | _SYSSMU11_2132424494\$ | PUBLIC | UNDO            | OFFLINE |
| 12         | _SYSSMU12_1803673755\$ | PUBLIC | UNDO            | OFFLINE |
| 13         | _SYSSMU13_3105490207\$ | PUBLIC | UNDO            | OFFLINE |
| 14         | _SYSSMU14_566209096\$  | PUBLIC | UNDO            | OFFLINE |
| 15         | _SYSSMU15_3645937391\$ | PUBLIC | UNDO            | OFFLINE |
| 16         | _SYSSMU16_522257056\$  | PUBLIC | UNDO            | OFFLINE |
| 17         | _SYSSMU17_1962353326\$ | PUBLIC | UNDO            | OFFLINE |
| 18         | _SYSSMU18_3333677753\$ | PUBLIC | UNDO            | OFFLINE |
| 19         | _SYSSMU19_3794987765\$ | PUBLIC | UNDO            | OFFLINE |
| 20         | _SYSSMU20_2419647899\$ | PUBLIC | UNDO            | OFFLINE |
| 21         | _SYSSMU21_3424807448\$ | PUBLIC | UNDO            | OFFLINE |

| SEGMENT_ID | SEGMENT_NAME           | OWNER  | TABLESPACE_NAME | STATUS  |
|------------|------------------------|--------|-----------------|---------|
| 22         | _SYSSMU22_3414158201\$ | PUBLIC | UNDO            | OFFLINE |
| 23         | _SYSSMU23_3571249787\$ | PUBLIC | UNDO            | OFFLINE |
| 24         | _SYSSMU24_568466869\$  | PUBLIC | UNDO            | OFFLINE |
| 25         | _SYSSMU25_3805201024\$ | PUBLIC | UNDO            | OFFLINE |
| 26         | _SYSSMU26_1031593024\$ | PUBLIC | UNDO            | OFFLINE |

27 rows selected.

```
SYS@orcl> alter system set undo_tablespace = undo;
```

System altered.

SYS@orcl> SYS@orcl> show parameter undo\_tablespace;

| NAME | TYPE | VALUE |
|------|------|-------|
|------|------|-------|

|                 |        |      |
|-----------------|--------|------|
| undo_tablespace | string | UNDO |
|-----------------|--------|------|

SYS@orcl> SYS@orcl> select segment\_id, segment\_name, owner, tablespace\_name, status from dba\_rollback\_segs;

| SEGMENT_ID | SEGMENT_NAME           | OWNER  | TABLESPACE_NAME | STATUS  |
|------------|------------------------|--------|-----------------|---------|
| 0          | SYSTEM                 | SYS    | SYSTEM          | ONLINE  |
| 1          | _SYSSMU1_1261223759\$  | PUBLIC | UNDOTBS1        | OFFLINE |
| 2          | _SYSSMU2_27624015\$    | PUBLIC | UNDOTBS1        | OFFLINE |
| 3          | _SYSSMU3_2421748942\$  | PUBLIC | UNDOTBS1        | OFFLINE |
| 4          | _SYSSMU4_625702278\$   | PUBLIC | UNDOTBS1        | OFFLINE |
| 5          | _SYSSMU5_2101348960\$  | PUBLIC | UNDOTBS1        | ONLINE  |
| 6          | _SYSSMU6_813816332\$   | PUBLIC | UNDOTBS1        | OFFLINE |
| 7          | _SYSSMU7_2329891355\$  | PUBLIC | UNDOTBS1        | OFFLINE |
| 8          | _SYSSMU8_399776867\$   | PUBLIC | UNDOTBS1        | OFFLINE |
| 9          | _SYSSMU9_1692468413\$  | PUBLIC | UNDOTBS1        | OFFLINE |
| 10         | _SYSSMU10_930580995\$  | PUBLIC | UNDOTBS1        | OFFLINE |
| 11         | _SYSSMU11_2132424494\$ | PUBLIC | UNDO            | ONLINE  |
| 12         | _SYSSMU12_1803673755\$ | PUBLIC | UNDO            | ONLINE  |
| 13         | _SYSSMU13_3105490207\$ | PUBLIC | UNDO            | ONLINE  |
| 14         | _SYSSMU14_566209096\$  | PUBLIC | UNDO            | ONLINE  |
| 15         | _SYSSMU15_3645937391\$ | PUBLIC | UNDO            | ONLINE  |
| 16         | _SYSSMU16_522257056\$  | PUBLIC | UNDO            | ONLINE  |
| 17         | _SYSSMU17_1962353326\$ | PUBLIC | UNDO            | ONLINE  |
| 18         | _SYSSMU18_3333677753\$ | PUBLIC | UNDO            | ONLINE  |
| 19         | _SYSSMU19_3794987765\$ | PUBLIC | UNDO            | ONLINE  |
| 20         | _SYSSMU20_2419647899\$ | PUBLIC | UNDO            | ONLINE  |
| 21         | _SYSSMU21_3424807448\$ | PUBLIC | UNDO            | ONLINE  |
| 22         | _SYSSMU22_3414158201\$ | PUBLIC | UNDO            | ONLINE  |
| 23         | _SYSSMU23_3571249787\$ | PUBLIC | UNDO            | ONLINE  |
| 24         | _SYSSMU24_568466869\$  | PUBLIC | UNDO            | ONLINE  |
| 25         | _SYSSMU25_3805201024\$ | PUBLIC | UNDO            | ONLINE  |
| 26         | _SYSSMU26_1031593024\$ | PUBLIC | UNDO            | ONLINE  |

27 rows selected.

```
SYS@orcl> select s.username, s.sid, s.serial#, t.xidusn undo_seg_num, r.name undo_seg_name, t.ubafil
undo_datafile_num, t.ubablk undo_block, t.used_ublk from v$session s, v$transaction t, v$rollname r where s.taddr
= t.addr and t.xidusn = r.usn;
```

| USERNAME              | SID        | SERIAL#   | UNDO_SEG_NUM | UNDO_SEG_NAME |
|-----------------------|------------|-----------|--------------|---------------|
| -----                 |            |           |              |               |
| - - - - -             |            |           |              |               |
| UNDO_DATAFILE_NUM     | UNDO_BLOCK | USED_UBLK |              |               |
| -----                 |            |           |              |               |
| HR                    | 10         | 53917     | 5            |               |
| _SYSSMU5_2101348960\$ |            |           |              |               |
| 4                     | 24325      | 1         |              |               |

- 트랜잭션이 살아있어 사용중인 undo tablespace의 segment는 online.

```
SYS@orcl> alter system kill session '10,53917' immediate;
System altered.
```

```
SYS@orcl> select segment_id, segment_name, owner, tablespace_name, status from dba_rollback_segs;
```

| SEGMENT_ID | SEGMENT_NAME           | OWNER  | TABLESPACE_NAME | STATUS  |
|------------|------------------------|--------|-----------------|---------|
| 0          | SYSTEM                 | SYS    | SYSTEM          | ONLINE  |
| 1          | _SYSSMU1_1261223759\$  | PUBLIC | UNDOTBS1        | OFFLINE |
| 2          | _SYSSMU2_27624015\$    | PUBLIC | UNDOTBS1        | OFFLINE |
| 3          | _SYSSMU3_2421748942\$  | PUBLIC | UNDOTBS1        | OFFLINE |
| 4          | _SYSSMU4_625702278\$   |        | PUBLIC UNDOTBS1 | OFFLINE |
| 5          | _SYSSMU5_2101348960\$  | PUBLIC | UNDOTBS1        | ONLINE  |
| 6          | _SYSSMU6_813816332\$   |        | PUBLIC UNDOTBS1 | OFFLINE |
| 7          | _SYSSMU7_2329891355\$  | PUBLIC | UNDOTBS1        | OFFLINE |
| 8          | _SYSSMU8_399776867\$   |        | PUBLIC UNDOTBS1 | OFFLINE |
| 9          | _SYSSMU9_1692468413\$  | PUBLIC | UNDOTBS1        | OFFLINE |
| 10         | _SYSSMU10_930580995\$  | PUBLIC | UNDOTBS1        | OFFLINE |
| 11         | _SYSSMU11_2132424494\$ | PUBLIC | UNDO            | ONLINE  |
| 12         | _SYSSMU12_1803673755\$ | PUBLIC | UNDO            | ONLINE  |
| 13         | _SYSSMU13_3105490207\$ | PUBLIC | UNDO            | ONLINE  |
| 14         | _SYSSMU14_566209096\$  | PUBLIC | UNDO            | ONLINE  |

|                           |             |        |
|---------------------------|-------------|--------|
| 15 _SYSSMU15_3645937391\$ | PUBLIC UNDO | ONLINE |
| 16 _SYSSMU16_522257056\$  | PUBLIC UNDO | ONLINE |
| 17 _SYSSMU17_1962353326\$ | PUBLIC UNDO | ONLINE |
| 18 _SYSSMU18_3333677753\$ | PUBLIC UNDO | ONLINE |
| 19 _SYSSMU19_3794987765\$ | PUBLIC UNDO | ONLINE |
| 20 _SYSSMU20_2419647899\$ | PUBLIC UNDO | ONLINE |
| 21 _SYSSMU21_3424807448\$ | PUBLIC UNDO | ONLINE |
| 22 _SYSSMU22_3414158201\$ | PUBLIC UNDO | ONLINE |
| 23 _SYSSMU23_3571249787\$ | PUBLIC UNDO | ONLINE |
| 24 _SYSSMU24_568466869\$  | PUBLIC UNDO | ONLINE |
| 25 _SYSSMU25_3805201024\$ | PUBLIC UNDO | ONLINE |
| 26 _SYSSMU26_1031593024\$ | PUBLIC UNDO | ONLINE |

27 rows selected.

- 트랜잭션을 kill하였으나 online으로 유지중이라 테이블스페이스 삭제할 수 없음.

SYS@orcl> drop tablespace undotbs1 including contents and datafiles;

Tablespace dropped.

SYS@orcl> select segment\_id, segment\_name, owner, tablespace\_name, status from dba\_rollback\_segs;

| SEGMENT_ID | SEGMENT_NAME           | OWNER  | TABLESPACE_NAME | STATUS |
|------------|------------------------|--------|-----------------|--------|
| -----      |                        |        |                 |        |
| 0          | SYSTEM                 | SYS    | SYSTEM          | ONLINE |
| 11         | _SYSSMU11_2132424494\$ | PUBLIC | UNDO            | ONLINE |
| 12         | _SYSSMU12_1803673755\$ | PUBLIC | UNDO            | ONLINE |
| 13         | _SYSSMU13_3105490207\$ | PUBLIC | UNDO            | ONLINE |
| 14         | _SYSSMU14_566209096\$  | PUBLIC | UNDO            | ONLINE |
| 15         | _SYSSMU15_3645937391\$ | PUBLIC | UNDO            | ONLINE |
| 16         | _SYSSMU16_522257056\$  | PUBLIC | UNDO            | ONLINE |
| 17         | _SYSSMU17_1962353326\$ | PUBLIC | UNDO            | ONLINE |
| 18         | _SYSSMU18_3333677753\$ | PUBLIC | UNDO            | ONLINE |
| 19         | _SYSSMU19_3794987765\$ | PUBLIC | UNDO            | ONLINE |
| 20         | _SYSSMU20_2419647899\$ | PUBLIC | UNDO            | ONLINE |
|            |                        |        |                 |        |
| SEGMENT_ID | SEGMENT_NAME           | OWNER  | TABLESPACE_NAME | STATUS |
| -----      |                        |        |                 |        |
| 21         | _SYSSMU21_3424807448\$ | PUBLIC | UNDO            | ONLINE |
| 22         | _SYSSMU22_3414158201\$ | PUBLIC | UNDO            | ONLINE |

|                           |             |        |
|---------------------------|-------------|--------|
| 23 _SYSSMU23_3571249787\$ | PUBLIC UNDO | ONLINE |
| 24 _SYSSMU24_568466869\$  | PUBLIC UNDO | ONLINE |
| 25 _SYSSMU25_3805201024\$ | PUBLIC UNDO | ONLINE |
| 26 _SYSSMU26_1031593024\$ | PUBLIC UNDO | ONLINE |

17 rows selected.

-- 삭제가 가능할 수 있음

-- 오류 발생하여 삭제 불가능할 경우

# 이전 언두 테이블스페이스에 속한 undo segment 중에 아직도 online상태로 되어 있는 경우는 테이블스페이스를 삭제 불가

# 해결방법

1. spfile을 이용해서 pfile생성

SYS@orcl> create pfile from spfile;

2. 데이터베이스를 정상적인 종료

SYS@orcl> shutdown immediate

3. pfile에 수동으로 undo segment를 offline 등록

SYS@orcl> ! vi \$ORACLE\_HOME/dbs/initiorcl.ora

SYS@orcl> \_offline\_rollback\_segments(\_SYSSMU2\_3151764486\$)

-- 여러개일 경우 , 사용

4. pfile로 시작

SYS@orcl> startup pfile=\$ORACLE\_HOME/dbs/initora19c.ora mount

SYS@orcl> select a.file#, b.name tbs\_name, a.name file\_name, a.checkpoint\_change#, a.status from v\$datafile a, v\$tablespace b where a.ts# = b.ts#;

| FILE#              | TBS_NAME                                   | FILE_NAME      |
|--------------------|--------------------------------------------|----------------|
| CHECKPOINT_CHANGE# | STATUS                                     |                |
| 1                  | /u01/app/oracle/oradata/ORCL/system01.dbf  | 4117793 SYSTEM |
| 2                  | /u01/app/oracle/oradata/ORCL/undotbs01.dbf | 4117793 ONLINE |
| 3                  | /u01/app/oracle/oradata/ORCL/sysaux01.dbf  | 4117793 ONLINE |
| 5                  | /u01/app/oracle/oradata/ORCL/undo01.dbf    | 4121809 ONLINE |

7 /u01/app/oracle/oradata/ORCL/users01.dbf

4117793 ONLINE

##### 5. offline 만들기

```
SYS@orcl> alter database datafile '/u01/app/oracle/oradata/ORCL/undotbs01.dbf' offline;
```

##### 6. open

```
SYS@orcl> alter database open
```

##### 7. 기존 테이블스페이스 삭제

```
SYS@orcl> drop tablespace undotbs1 including contents and datafiles;
```

##### # 원래대로 돌리기

```
SYS@orcl> create undo tablespace undotbs1 datafile '/u01/app/oracle/oradata/ORCL/undotbs01' size 50m  
autoextend on;
```

Tablespace created.

```
SYS@orcl> alter system set undo_tablespace = undotbs1;
```

System altered.

```
SYS@orcl> drop tablespace undo including contents and datafiles;
```

Tablespace dropped.

```
SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a,  
v$tablespace b where a.ts# = b.ts#;
```

| FILE# | TBS_NAME | FILE_NAME                                 | CHECKPOINT_CHANGE# |
|-------|----------|-------------------------------------------|--------------------|
| 1     | SYSTEM   | /u01/app/oracle/oradata/ORCL/system01.dbf | 2595390            |
| 2     | UNDOTBS1 | /u01/app/oracle/oradata/ORCL/undotbs01    | 2597795 ONLINE     |
| 3     | SYSAUX   | /u01/app/oracle/oradata/ORCL/sysaux01.dbf | 2595390            |
| 7     | USERS    | /u01/app/oracle/oradata/ORCL/users01.dbf  | 2595390            |

## &lt;&lt;시나리오13&gt;&gt; 복구 작업시 필요한 archive log file이 한쪽에 없을 경우

```
SYS@orcl> archive log list
```

```
Database log mode           Archive Mode
Automatic archival          Enabled
Archive destination         /home/oracle/arch2
Oldest online log sequence  66
Next log sequence to archive 68
Current log sequence        68
```

```
SYS@orcl> !ls /home/oracle/arch*
```

```
/home/oracle/arch1:
```

```
0001_0000000062_1199981728.arc 0001_0000000064_1199981728.arc 0001_0000000066_1199981728.arc
0001_0000000063_1199981728.arc 0001_0000000065_1199981728.arc 0001_0000000067_1199981728.arc
```

```
/home/oracle/arch2:
```

```
0001_0000000062_1199981728.arc 0001_0000000064_1199981728.arc 0001_0000000066_1199981728.arc
0001_0000000063_1199981728.arc 0001_0000000065_1199981728.arc 0001_0000000067_1199981728.arc
```

```
# 장애 발생
```

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/users01.dbf
```

```
SYS@orcl> !ls /home/oracle/arch*
```

```
/home/oracle/arch1:
```

```
0001_0000000062_1199981728.arc 0001_0000000064_1199981728.arc 0001_0000000066_1199981728.arc
0001_0000000063_1199981728.arc 0001_0000000065_1199981728.arc 0001_0000000067_1199981728.arc
```

```
/home/oracle/arch2:
```

```
0001_0000000062_1199981728.arc 0001_0000000064_1199981728.arc 0001_0000000066_1199981728.arc
0001_0000000063_1199981728.arc 0001_0000000065_1199981728.arc 0001_0000000067_1199981728.arc
```

```
SYS@orcl> !rm /home/oracle/arch2/0001_0000000064_1199981728.arc
```

```
# Recovery
```

```
1. 복구 수행해야할 테이블스페이스를 offline immediate
```

```
SYS@orcl> shut abort
```

```
ORACLE instance shut down.
```

```
SYS@orcl> startup
```

ORACLE instance started.

Total System Global Area 1811936368 bytes  
Fixed Size 8897648 bytes  
Variable Size 436207616 bytes  
Database Buffers 1358954496 bytes  
Redo Buffers 7876608 bytes  
Database mounted.

ORA-01157: cannot identify/lock data file 7 - see DBWR trace file

ORA-01110: data file 7: '/u01/app/oracle/oradata/ORCL/users01.dbf'

SYS@orcl> alter database datafile 7 offline;  
Database altered.

SYS@orcl> select a.file#, b.name tbs\_name, a.name file\_name, a.checkpoint\_change#, a.status from v\$datafile a,  
v\$tablespace b where a.ts# = b.ts#;

| FILE# | TBS_NAME | FILE_NAME                                 | CHECKPOINT_CHANGE# |
|-------|----------|-------------------------------------------|--------------------|
| 1     | SYSTEM   | /u01/app/oracle/oradata/ORCL/system01.dbf | 2595390            |
| 3     | SYSAUX   | /u01/app/oracle/oradata/ORCL/sysaux01.dbf | 2595390            |
| 2     | UNDOTBS1 | /u01/app/oracle/oradata/ORCL/undotbs01    | 2597795            |
| 7     | USERS    | /u01/app/oracle/oradata/ORCL/users01.dbf  | 2595390            |

2. 최근 백업본을 찾아 restore

SYS@orcl> ! cp -v /home/oracle/backup/open\_bkp/users01.dbf /u01/app/oracle/oradata/ORCL/  
'/home/oracle/backup/open\_bkp/users01.dbf' -> '/u01/app/oracle/oradata/ORCL/users01.dbf'

3. 백업 이후에 변경정보(리두) 적용

SYS@orcl> recover tablespace users

ORA-00279: change 2192042 generated at 05/12/2025 18:43:03 needed for thread 1

ORA-00289: suggestion : /home/oracle/arch2/0001\_0000000062\_1199981728.arc

ORA-00280: change 2192042 for thread 1 is in sequence #62

Specify log: {<RET>=suggested | filename | AUTO | CANCEL}  
/home/oracle/arch2/0001\_0000000062\_1199981728.arc  
ORA-00279: change 2392382 generated at 05/12/2025 19:24:05 needed for thread 1  
ORA-00289: suggestion : /home/oracle/arch2/0001\_0000000063\_1199981728.arc  
ORA-00280: change 2392382 for thread 1 is in sequence #63

Specify log: {<RET>=suggested | filename | AUTO | CANCEL}  
/home/oracle/arch2/0001\_0000000063\_1199981728.arc  
ORA-00279: change 2594334 generated at 05/12/2025 19:41:43 needed for thread 1  
ORA-00289: suggestion : /home/oracle/arch2/0001\_0000000064\_1199981728.arc  
ORA-00280: change 2594334 for thread 1 is in sequence #64

Specify log: {<RET>=suggested | filename | AUTO | CANCEL}  
/home/oracle/arch1/0001\_0000000064\_1199981728.arc  
ORA-00279: change 2595377 generated at 05/12/2025 19:51:30 needed for thread 1  
ORA-00289: suggestion : /home/oracle/arch2/0001\_0000000065\_1199981728.arc  
ORA-00280: change 2595377 for thread 1 is in sequence #65

Specify log: {<RET>=suggested | filename | AUTO | CANCEL}  
/home/oracle/arch2/0001\_0000000065\_1199981728.arc  
ORA-00279: change 2595380 generated at 05/12/2025 19:51:32 needed for thread 1  
ORA-00289: suggestion : /home/oracle/arch2/0001\_0000000066\_1199981728.arc  
ORA-00280: change 2595380 for thread 1 is in sequence #66

Specify log: {<RET>=suggested | filename | AUTO | CANCEL}  
/home/oracle/arch2/0001\_0000000066\_1199981728.arc  
Log applied.  
Media recovery complete.

#### 4. 복원된 테이블스페이스 online

SYS@orcl> alter tablespace users online;  
Tablespace altered.